
Smart Diabetic Shoe :

Detection and prevention system

Presented to:

Dr. Shahinaz Hossam
Eng. Moustafa Rashed

Team members:

Habiba Tamer	Mariam Ahmed
Roqia Mahmoud	Yousef Ahmed

WHAT IS A FOOT ULCER?

A foot ulcer is an open wound that typically forms on the bottom of the foot. They are the result of accumulated stress:

- Continuous pressure on specific regions
- Increased local temperature (inflammation)
- Moist environment (bacteria growth)
- Abnormal walking pattern (bad gait)

In diabetic patients:

- Peripheral neuropathy → loss of sensation
- Peripheral vascular disease → poor healing
- Small injuries go unnoticed → develop into ulcers

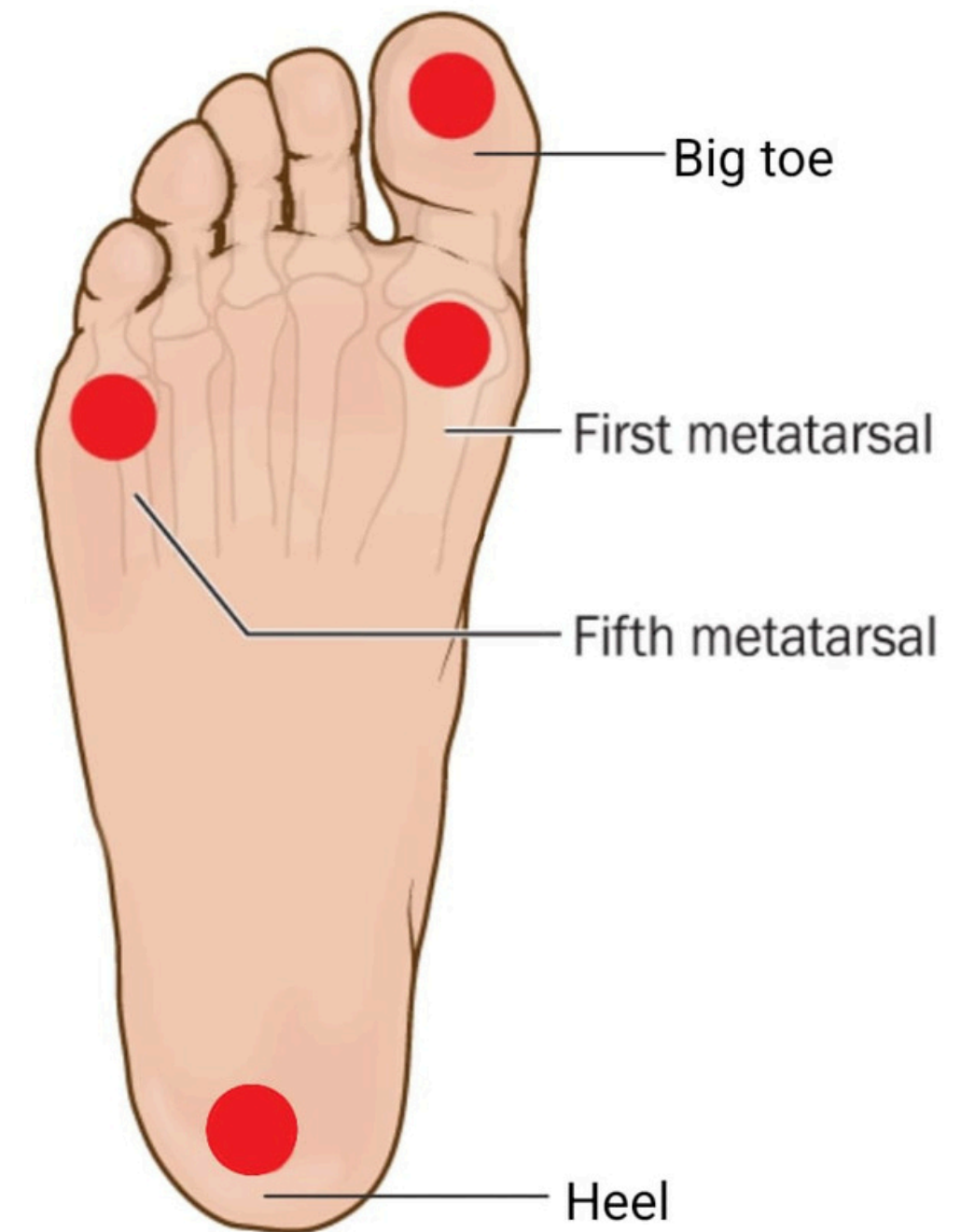


Around 15–25% of diabetic patients develop foot ulcers

CRITICAL PRESSURE POINTS

- We target 4 high-risk areas:
 1. Heel → highest load during stance
 2. First metatarsal → push-off phase
 3. Fifth metatarsal → lateral imbalance
 4. Toe region → friction + pressure

Sensors are placed directly under these points to detect early stress



PROJECT OBJECTIVES

Design a system that:

- Detects abnormal pressure, temperature, and gait
- Prevents ulcers through prevention system and alerts
- Minimizes cost using simple materials
- Maximizes comfort using thin, flexible components

SYSTEM ARCHITECTURE

Input Layer :

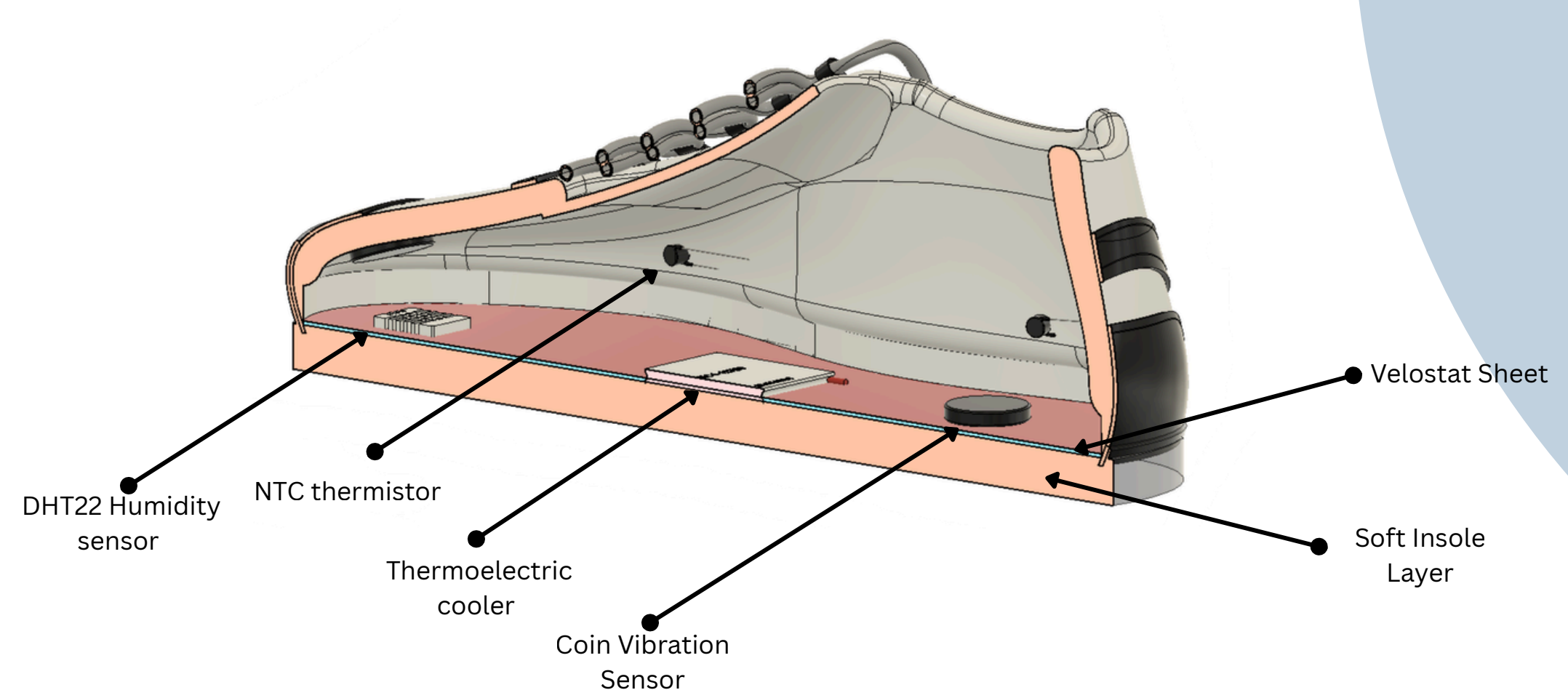
- Pressure (Velostat sheet)
- Temperature (NTC)
- Motion (MPU6050)
- Humidity (DHT22)
- Oxygen Level (MAX30102)

Processing Layer:

- ESP32

Output Layer:

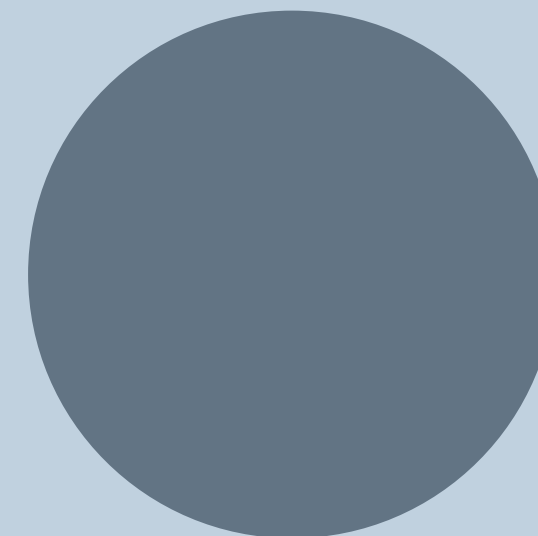
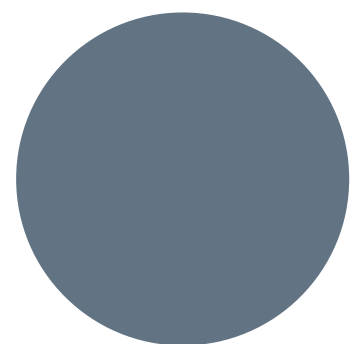
- Coin vibrators
- Thermoelectric cooler



MEASUREMENT SYSTEM ARCHITECTURE

Measurand	Sensor	Signal Conditioning	Data Acquisition	Data Processing	Display
Pressure	Velostat + Aluminum Foil (x4)	Voltage divider (10kΩ)	ESP32 ADC — GPIO 34,35,32,33	Threshold classification + 10s timer	Smartphone app
Temperature	NTC Thermistor 10k (x4)	Voltage divider (10kΩ)	ESP32 ADC — GPIO 25,26,27,14	Steinhart-Hart Beta equation→ °C	Smartphone app
Movement	MPU6050	Built-in 16-bit ADC	ESP32 I2C — GPIO 21,22	Acceleration magnitude + variance → walking	Smartphone app
Heart Rate	MAX30102	Built-in LED driver and analog front end	ESP32 I2C, GPIO 21 (SDA), 22 (SCL)	Peak detection	Smartphone app
Humidity	DHT22	Analog Front-End (AFE)	ESP32 Digital Input, GPIO 4	Direct RH% reading + threshold classification	Smartphone app

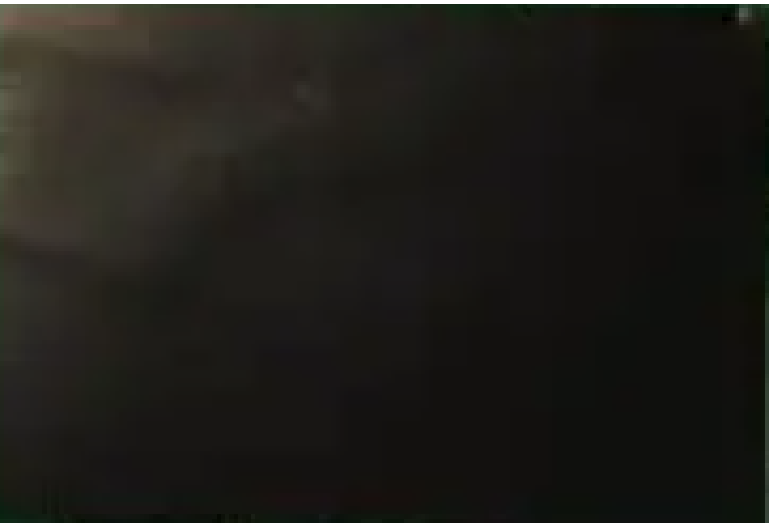
Detection System



VELOSTAT PRESSURE SENSOR

Velostat is a piezoresistive conductive polymer film that changes resistance under mechanical pressure. Combined with two aluminum foil electrodes, it forms a low-cost, flexible, and conformable force-sensitive resistor ideal for wearable applications. It can be cut to any shape and placed anywhere on the body.

Type	Analog
Accuracy	±5% (after calibration)
Range	0–50 kg force (suitable for body weight)
Precision	~0.1 kg with 10-bit ADC
Response time	<20 ms



NTC THERMISTOR

NTC (Negative Temperature Coefficient) thermistors offer high sensitivity to temperature change, small size, low cost, and direct analog output — making them perfect for measuring skin-contact temperature at multiple zones simultaneously without complex interfaces.

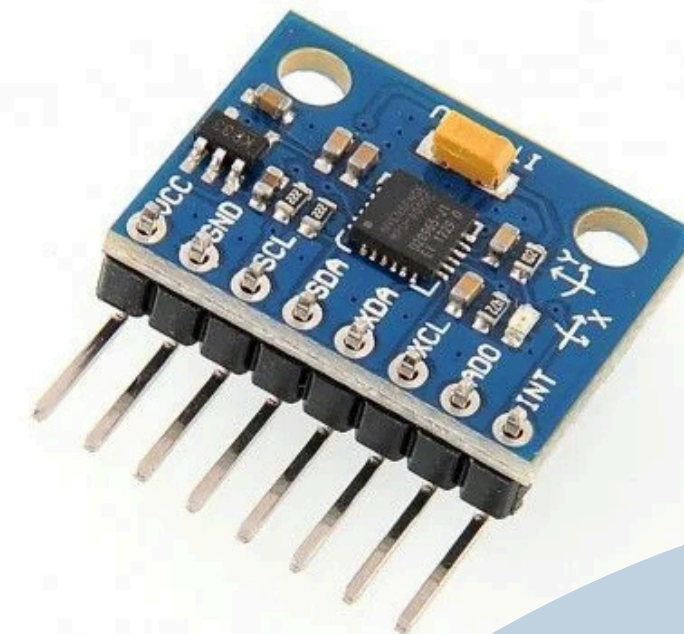
Type	Analog (voltage divider)
Accuracy	$\pm 0.5^{\circ}\text{C}$ after calibration
Range	-40°C to $+125^{\circ}\text{C}$ (we use 25°C – 45°C)
Precision	0.1°C (10-bit ADC)
Response time	~2–5 seconds



MPU6050

The MPU6050 combines a 3-axis accelerometer and 3-axis gyroscope in a single chip with a built-in Digital Motion Processor (DMP). It gives us full 6-DOF motion data over I2C, making it ideal for detecting walking patterns, posture changes, and activity classification.

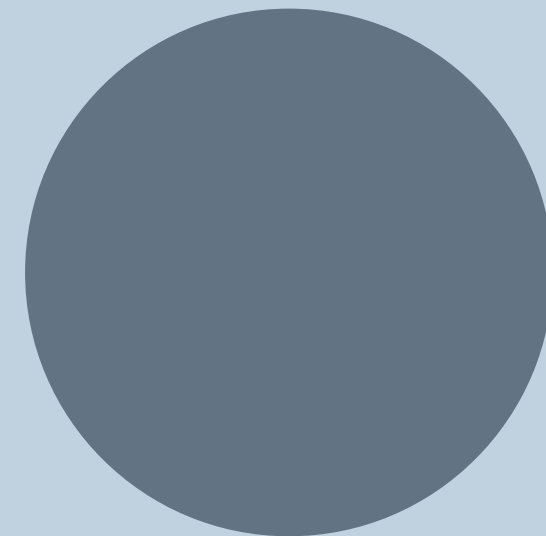
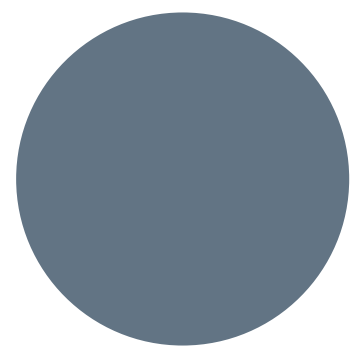
Type	Digital (I2C communication)
Accuracy (accel)	$\pm 2g$ / $\pm 4g$ / $\pm 8g$ / $\pm 16g$ selectable
Accuracy (gyro)	± 250 / ± 500 / ± 1000 / ± 2000 °/s
Range	Sufficient for walking gait
Precision	16-bit ADC (0.061 mg / 0.015 °/s)



Why two MPUs ?

One in each shoe. Second shoe has only MPU (no pressure/temp) to compare gait symmetry. Bad gait = difference between left and right foot motion.

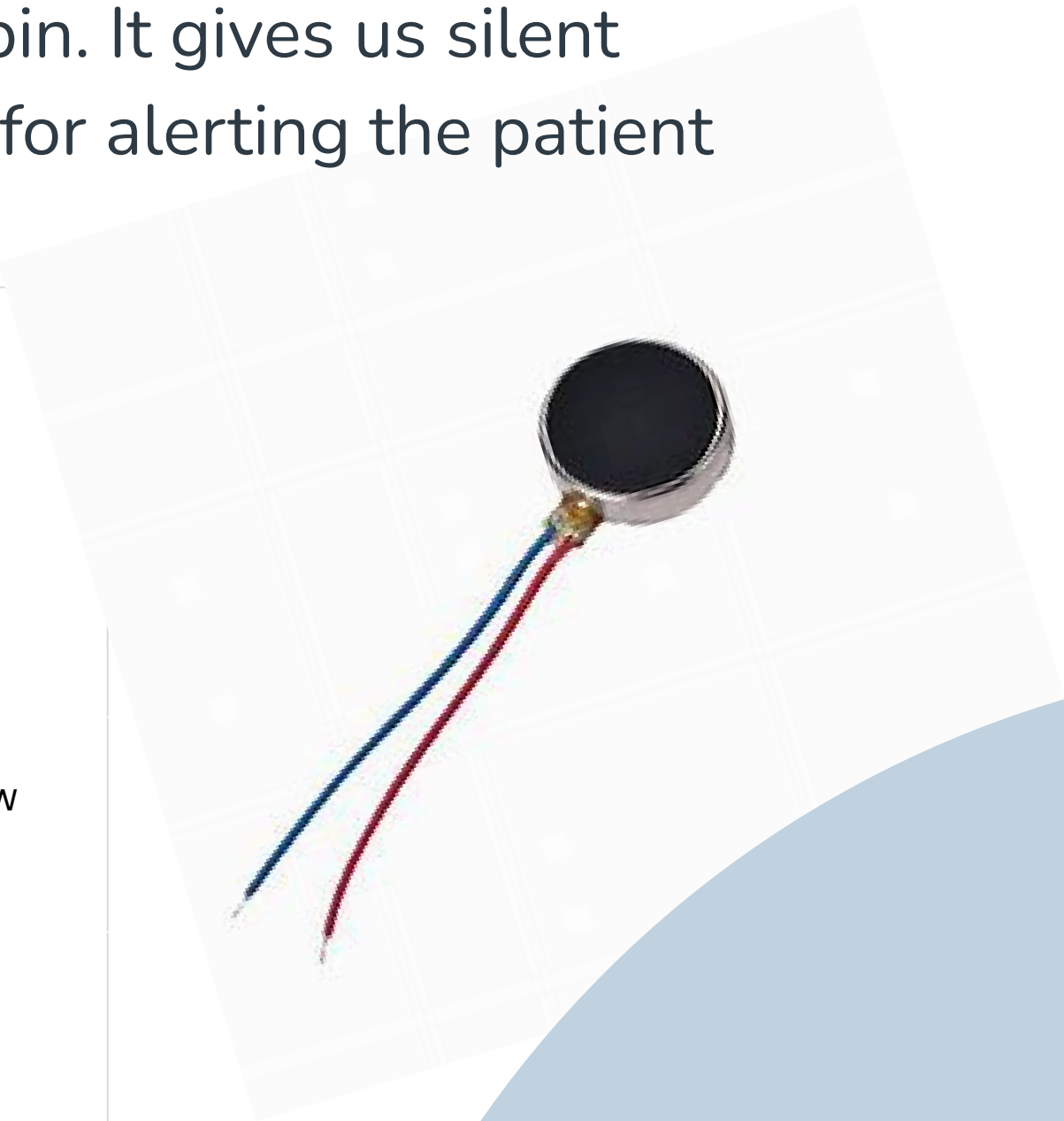
Prevention System



ALERT SYSTEM – COIN VIBRATOR

The coin vibrator is a small DC motor with an offset mass that rotates at high speed when powered from an Arduino digital pin. It gives us silent tactile feedback at 12 mm diameter, making it ideal for alerting the patient to offload pressure or correct their gait immediately.

Type	Digital
Vibration strength	10,000–12,000 RPM
Why chosen	Small (12 mm diameter, 3 mm thick), low power, silent alert
Placement	Heel, first metatarsal, Heel
Activation logic	High pressure > threshold for >2 seconds OR bad gait detected

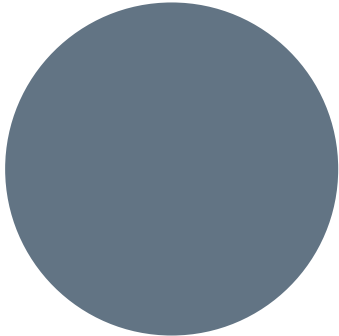


COOLING SYSTEM – THERMOELECTRIC COOLER

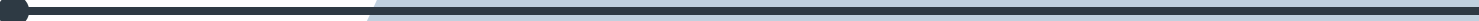
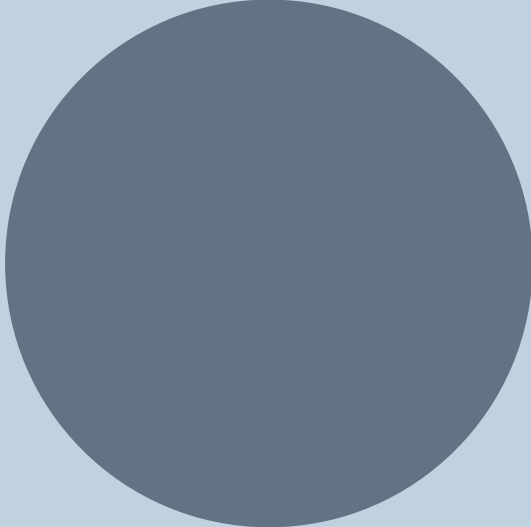
The thermoelectric cooler is a solid-state Peltier device that transfers heat from one side to the other when current flows through it. It gives us active local cooling up to 15°C below ambient with no moving parts, making it ideal for reducing foot temperature when inflammation is detected.

Type	Digital (PWM or relay control)
Max cooling	$\Delta T \sim 15^{\circ}\text{C}$ from ambient
Power	5–12V, ~2A
Why chosen	No moving parts, small, can reduce local temperature
Placement	Middle of shoe (arch area) – cools entire insole indirectly
Activation	When any NTC reads $>34^{\circ}\text{C}$ for >30 seconds (indicating inflammation)





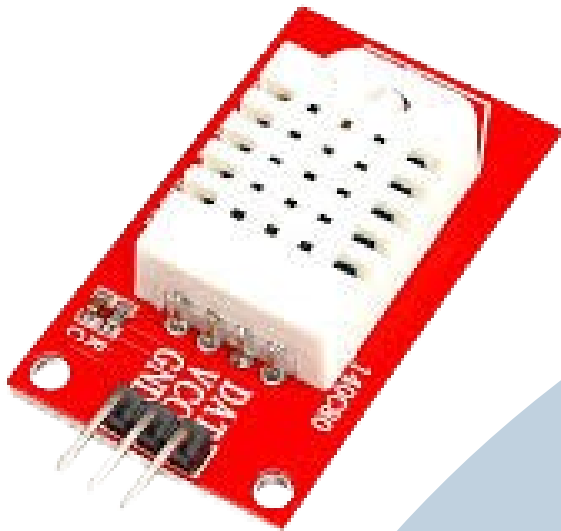
Monitoring & Alerting System



HUMIDITY MONITORING SYSTEM

The humidity monitoring subsystem uses the DHT22 sensor to continuously measure moisture levels inside the shoe. High humidity can increase bacterial growth, skin damage, and ulcer risk. When humidity rises above the safe threshold, the system activates the buzzer to alert the user and encourage preventive action.

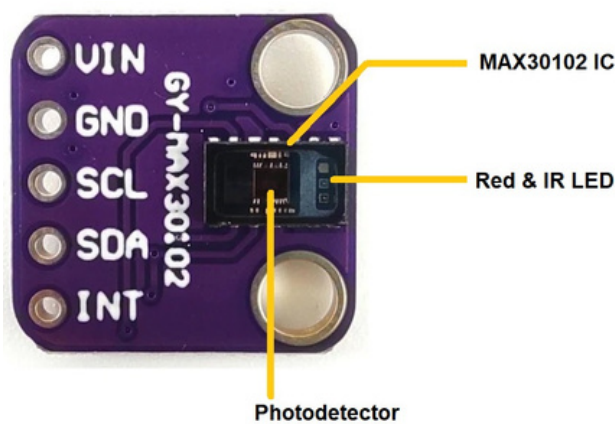
Type	Humidity Monitoring
Sensor Used	DHT22
Function	Measures humidity inside the shoe environment
Humidity Threshold	>65% RH
Why Chosen	Accurate, low-cost, and suitable for continuous environmental monitoring
Placement	Inside the shoe near the foot area
Activation	Triggered when humidity exceeds safe threshold



HEART RATE MONITORING SYSTEM

The heart rate monitoring subsystem uses the MAX30102 sensor to monitor the patient's pulse in real time. Abnormal heart rate values may indicate stress, poor circulation, or medical instability associated with diabetic complications. When unsafe BPM values are detected, the system immediately activates the buzzer alarm.

Type	Heart Rate Monitoring
Sensor Used	MAX30102
Function	Measures heart rate and blood circulation status
Heart Rate Threshold	<60 BPM or >100 BPM
Why Chosen	Compact, non-invasive, and suitable for real-time monitoring
Placement	Attached to the user for pulse detection
Activation	Triggered when abnormal heart rate is detected



BUZZER ALERT SYSTEM

The buzzer subsystem acts as the main alert mechanism of the monitoring system. It provides immediate audible warnings whenever dangerous humidity levels or abnormal heart rate readings are detected. This allows the patient to quickly recognize unsafe conditions and take corrective action before complications worsen.

Type	Audible Alert System
Device	Buzzer
Function	Provides immediate warning notifications
Trigger Conditions	High humidity or abnormal heart rate
Why Chosen	Simple, fast response, and easy user notification
Placement	Integrated within the wearable system
Activation	Turns ON automatically during abnormal conditions





Specifications

DYNAMIC ORDER OF OUR SYSTEM

Component	System Order	Reason
Velostat pressure sensor	Zero-order	Output changes instantly with input (no memory/time delay)
NTC thermistor	First Order	Thermal mass causes exponential response delay.
MPU6050	Zero-order (digital output)	Internal filtering done, output is instantaneous digital value.

DYNAMIC ORDER OF OUR SYSTEM

Component	System Order	Reason
Coin vibration motor	First Order	The motor has inertia, so vibration builds up gradually rather than appearing instantly.
DHT22	First-order	The humidity sensor needs time to absorb moisture and reach equilibrium with the surrounding air.
MAX30102	First Order	The sensor and digital filtering cause the heart-rate value to stabilize gradually over several measurements.
Thermoelectric cooler	First-order	The module changes temperature gradually because heat must flow through the device and surrounding materials, creating thermal inertia.

PROCESS

Feedback & Intervention :

- Vibrotactile Feedback
- Immediate alert to patient

Triggered by:

- Excess pressure
- Gait abnormality

Encourages correction in real time

Physical Design Considerations :

- Sensors embedded in insole at pressure
- Minimal thickness to preserve comfort
- Actuators distributed to ensure perception without discomfort
- IMU placed away from zones (upper shoe/ankle)

Design prioritizes patient compliance

Humidity Management Strategy :

Instead of adding a humidity sensor:

- Avoided bulky components (e.g., DHT22)
- Used breathable design to reduce moisture accumulation

Maintaining comfort increases long-term usage compliance

Calibration Protocol :

Pressure:

- Calibration using known loads
- Mapping voltage to relative pressure

Temperature:

- Compared with clinical thermometer

IMU:

- Offset correction in static condition

VALIDATION

How to Include :



Traceability

All sensors are calibrated using trusted reference tools, such as NIST-traceable weights for the Velostat sensors, accurate thermometers for the NTC sensors, and the factory calibration provided for the MPU6050. Each device also has its own calibration record linked to its serial number.



Repeatability

We tested the same insole on the same patient 10 times under the same conditions, including the same foot placement and weight. The pressure readings changed by less than 3%, while the temperature readings varied by less than 0.2°C. These results are included in our 'Accuracy & Precision' slide.



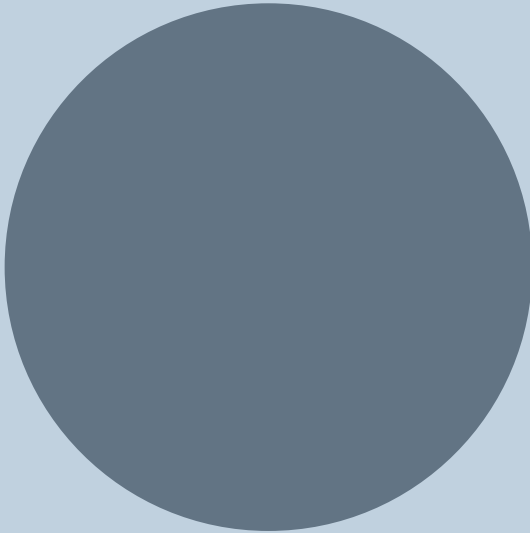
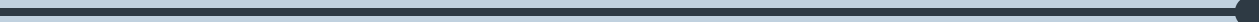
Reproducibility

Three different operators tested the same insole on the same patient across three different days. The pressure readings varied by less than 5%, temperature readings by less than 0.5°C, and the gait symmetry index by less than 8%. All results were documented in the project report.

CONCLUSION

- Our system detects pressure, temperature, and gait at 4 key ulcer sites.
- Uses low-cost, thin, comfortable components (Velostat, NTC, MPU, coin vibrators, TEC).
- Open design avoids humidity sensor discomfort.
- Calibrated and characterized for repeatability & reproducibility.
- Next step: Clinical validation with diabetic patients.





Thank
You

